

| Hands-on Activity 5.1: Implement Ansible roles in playbooks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                            |
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| <b>Course/Section:</b> CPE 212-CPE31S1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Date Submitted:</b> January 15, 2026    |
| <b>Instructor:</b> Engr. Robin Valenzuela                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Semester and SY:</b> 3rd Year - 2nd Sem |
| Activity 5: Consolidating Playbook plays                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                            |
| <b>1. Objectives:</b><br>1.1 Use <b>when</b> command in playbook for different OS distributions<br>1.2 Apply refactoring techniques in cleaning up the playbook codes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                            |
| <b>2. Discussion:</b><br><br><p>We are going to look at a way that we can differentiate a playbook by a host in terms of which distribution the host is running. It's very common in most Linux shops to run multiple distributions, for example, Ubuntu shop or Debian shop and you need a different distribution for a one off-case or perhaps you want to run plays only on certain distributions.</p> <p>It is a best practice in ansible when you are working in a collaborative environment to use the command git pull. git pull is a Git command used to update the local version of a repository from a remote. By default, git pull does two things. Updates the current local working branch (currently checked out branch) and updates the remote-tracking branches for all other branches. git pull essentially pulls down any changes that may have happened since the last time you worked on the repository.</p> <p><b>Requirement:</b><br/> In this activity, you will need to create a CentOS VM. Likewise, you need to activate the second adapter to a host-only adapter after the installations. Take note of the IP address of the CentOS VM. Make sure to use the command <b>ssh-copy-id</b> to copy the public key to CentOS. Verify if you can successfully SSH to CentOS VM.</p> |                                            |
| <b>Task 1: Use when command for different distributions</b><br><br>1. In the local machine, make sure you are in the local repository directory ( <b>CPE232_yourname</b> ). Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issued this command. Did something happen? Why?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |

```

jalene@workstation:~/CPE212_jalene$ git pull
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 739 bytes | 369.00 KiB/s, done.
From github.com:jaliebee/CPE212_jalene
   68fbbe2..fb1d79b  main       -> origin/main
hint: You have divergent branches and need to specify how to reconcile them.
hint: You can do so by running one of the following commands sometime before
hint: your next pull:
hint:
hint:   git config pull.rebase false  # merge
hint:   git config pull.rebase true   # rebase
hint:   git config pull.ff only       # fast-forward only
hint:
hint: You can replace "git config" with "git config --global" to set a default
hint: preference for all repositories. You can also pass --rebase, --no-rebase,
hint: or --ff-only on the command line to override the configured default per
hint: invocation.
fatal: Need to specify how to reconcile divergent branches.
jalene@workstation:~/CPE212_jalene$ ls
ansible.cfg  install_apache.yml  inventory.ini  playbook.yml  README.md
jalene@workstation:~/CPE212_jalene$

```

Basically, I can perform git pull but the Git stopped because my local branch and remote branch have repository changes. The Git is asking whether I want to merge, rebase, or only fast-forward.

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): ***ansible-playbook --ask-become-pass install\_apache.yml***. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the package manager. The default package manager for Centos is "yum."
3. Edit the ***install\_apache.yml*** file and insert the lines shown below.

```
---
- hosts: all
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes
        when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      apt:
        name: apache2
        when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
        when: ansible_distribution == "Ubuntu"
```

Make sure to save the file and exit.

Run *ansible-playbook --ask-become-pass install\_apache.yml* and describe the result.

If you have a mix of Debian and Ubuntu servers, you can change the configuration of your playbook like this.

```
- name: update repository index
  apt:
    update_cache: yes
    when: ansible_distribution in ["Debian", "Ubuntu"]
```

*Note:* This will work also if you try. Notice the changes are highlighted.

4. Edit the *install\_apache.yml* file and insert the lines shown below.

```

---
- hosts: all
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes
        when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      apt:
        name: apache2
        state: latest
        when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
        state: latest
        when: ansible_distribution == "Ubuntu"

    - name: update repository index
      dnf:
        update_cache: yes
        when: ansible_distribution == "CentOS"

    - name: install apache2 package
      dnf:
        name: httpd
        state: latest
        when: ansible_distribution == "CentOS"

    - name: add PHP support for apache
      dnf:
        name: php
        state: latest
        when: ansible_distribution == "CentOS"

```

Make sure to save and exit.

Run *ansible-playbook --ask-become-pass install\_apache.yml* and describe the result.

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in the CentOS is not yet active. Thus, you need to activate it first.

5.1 To activate, go to the CentOS VM terminal and enter the following:

*systemctl status httpd*

The result of this command tells you that the service is inactive.

5.2 Issue the following command to start the service:

*sudo systemctl start httpd*

(When prompted, enter the sudo password)

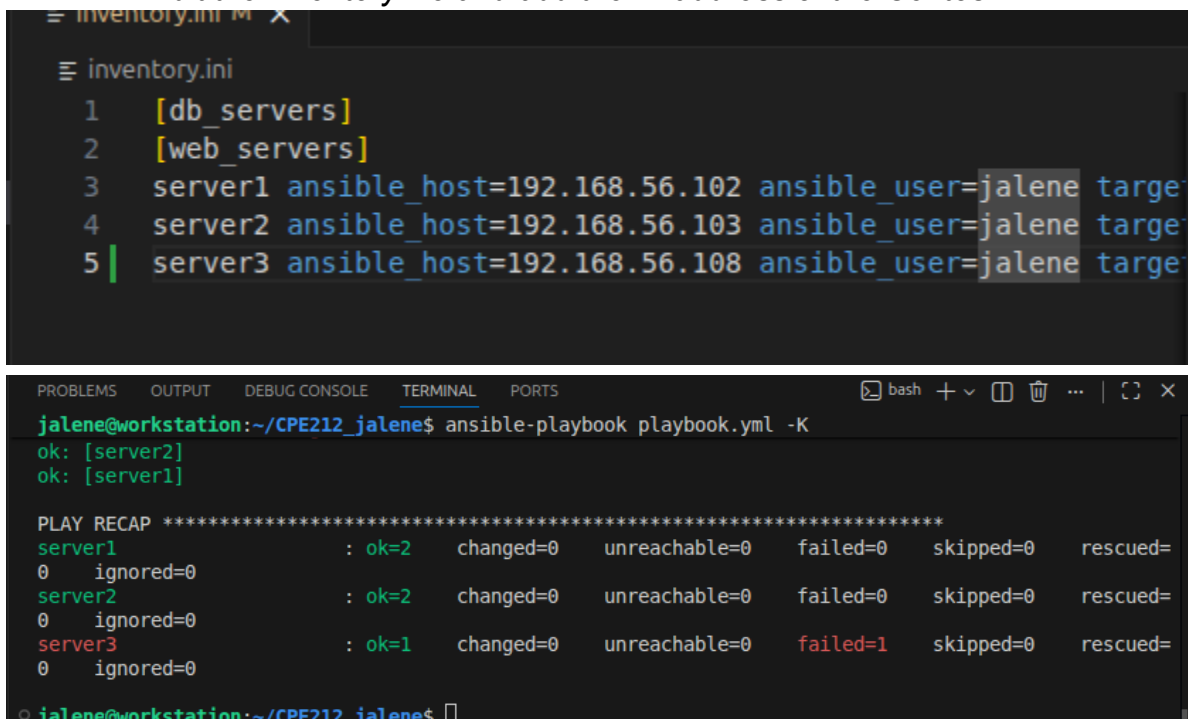
*sudo firewall-cmd --add-port=80/tcp*

(The result should be a success)

5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)

### Task 1

*Edit the inventory file and add the IP address of the Centos VM*



```
= inventory.ini M X
inventory.ini
1  [db_servers]
2  [web_servers]
3  server1 ansible_host=192.168.56.102 ansible_user=jalene target=
4  server2 ansible_host=192.168.56.103 ansible_user=jalene target=
5  server3 ansible_host=192.168.56.108 ansible_user=jalene target=

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
jalene@workstation:~/CPE212_jalene$ ansible-playbook playbook.yml -K
ok: [server2]
ok: [server1]

PLAY RECAP *****
server1 : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=
0 ignored=0
server2 : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=
0 ignored=0
server3 : ok=1 changed=0 unreachable=0 failed=1 skipped=0 rescued=
0 ignored=0
jalene@workstation:~/CPE212_jalene$
```

*Edit the install\_apache.yml file*

The screenshot shows a VS Code editor with a file named `install_apache.yml` open. The file contains an Ansible playbook with the following tasks:

```
! install_apache.yml
1 ---
2 - hosts: all
3   become: true
4   tasks:
5
6     - name: update repository index
7       apt:
8         update_cache: yes
9       when: ansible_distribution == "Ubuntu"
10
11    - name: install apache2 package
12      apt:
13        name: apache2
14      when: ansible_distribution == "Ubuntu"
15
16    - name: add PHP support for apache
17      apt:
18        name: libapache2-mod-php
19      when: ansible_distribution == "Ubuntu"
20
```

Below the editor, the terminal shows the output of the command `ansible-playbook install_apache.yml -K`:

```
jalene@workstation:~/CPE212_jalene$ ansible-playbook install_apache.yml -K
server1      : ok=4  changed=1  unreachable=0  failed=0  skipped=3  rescued=
0 ignored=0
server2      : ok=4  changed=1  unreachable=0  failed=0  skipped=3  rescued=
0 ignored=0
server3      : ok=1  changed=0  unreachable=0  failed=1  skipped=3  rescued=
0 ignored=0
jalene@workstation:~/CPE212_jalene$
```

*Edit the `install_apache.yml` file*

The screenshot shows the same VS Code editor with the `install_apache.yml` file, but it has been edited to use `yum` instead of `apt` for CentOS:

```
! install_apache.yml
18     name: libapache2-mod-php
19     when: ansible_distribution == "Ubuntu"
20
21    - name: update repository index
22      yum:
23        update_cache: yes
24      when: ansible_distribution == "CentOS"
25
26    - name: install apache2 package
27      yum:
28        name: httpd
29        state: latest
30      when: ansible_distribution == "CentOS"
```

The terminal output shows the command `ansible-playbook install_apache.yml -K` being executed. The tasks for `server1` and `server2` are skipped because they are Ubuntu servers. The task for `server3` is executed successfully because it is a CentOS server.

```
jalene@workstation:~/CPE212_jalene$ ansible-playbook install_apache.yml -K
TASK [install apache2 package] *****
skipping: [server1]
skipping: [server2]

TASK [add PHP support for apache] *****
skipping: [server1]
skipping: [server2]

PLAY RECAP *****
server1      : ok=4  changed=1  unreachable=0  failed=0  skipped=3  rescued=
0 ignored=0
server2      : ok=4  changed=1  unreachable=0  failed=0  skipped=3  rescued=
0 ignored=0
server3      : ok=1  changed=0  unreachable=0  failed=1  skipped=3  rescued=
0 ignored=0
jalene@workstation:~/CPE212_jalene$
```

```
[jalene@server3 ~]$ sudo dnf install httpd -y
[sudo] password for jalene:
CentOS Stream 9 - BaseOS                                0.0 B/s | 0 B    00:00
Errors during downloading metadata for repository 'baseos':
 - Curl error (6): Couldn't resolve host name for https://mirrors.centos.org/metalink?repo=centos-baseos-9-stream&arch=x86_64&protocol=https,http [Could not resolve host: mirrors.centos.org]
Error: Failed to download metadata for repo 'baseos': Cannot prepare internal mirrorlist: Curl error (6): Couldn't resolve host name for https://mirrors.centos.org/metalink?repo=centos-baseos-9-stream&arch=x86_64&protocol=https,http [Could not resolve host: mirrors.centos.org]
[jalene@server3 ~]$
```

*This is the proof why I can't perform the tasks for CentOS. This is the same output shown whenever I try to run the ansible with my CentOS server.*

## Task 2: Refactoring playbook

This time, we want to make sure that our playbook is efficient and that the codes are easier to read. This will also make run ansible more quickly if it has to execute fewer tasks to do the same thing.

1. Edit the playbook *install\_apache.yml*. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

```

---
- hosts: all
  become: true
  tasks:

    - name: update repository index Ubuntu
      apt:
        update_cache: yes
        when: ansible_distribution == "Ubuntu"

    - name: install apache2 and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        when: ansible_distribution == "Ubuntu"

    - name: update repository index for CentOS
      dnf:
        update_cache: yes
        when: ansible_distribution == "CentOS"

    - name: install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
        when: ansible_distribution == "CentOS"

```

Make sure to save the file and exit.

Run *ansible-playbook --ask-become-pass install\_apache.yml* and describe the result.

2. Edit the playbook *install\_apache.yml* again. In task 2.1, we consolidated the plays into one play. This time we can actually consolidate everything in just 2 plays. This can be done by removing the update repository play and putting the command *update\_cache: yes* below the command *state: latest*. See below for reference:

```

---
- hosts: all
  become: true
  tasks:

    - name: install apache2 and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes
      when: ansible_distribution == "CentOS"

```

Make sure to save the file and exit.

Run *ansible-playbook --ask-become-pass install\_apache.yml* and describe the result.

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the `apache_package` and `php_package` are variables. The names are arbitrary, which means we can choose different names. We also take out the line `when: ansible_distribution`. Edit the playbook *install\_apache.yml* again and make sure to follow the below image. Make sure to save the file and exit.

```

---
- hosts: all
  become: true
  tasks:

    - name: install apache and php
      apt:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes

```

Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the result.

- Unfortunately, task 2.3 was not successful. It's because we need to change something in the inventory file so that the variables we declared will be in place. Edit the `inventory` file and follow the below configuration:

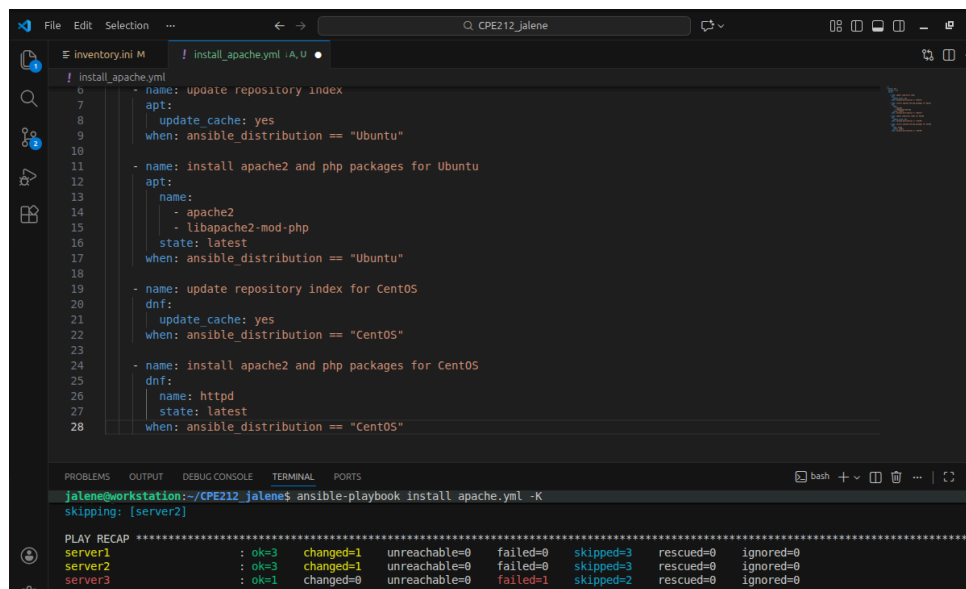
```
192.168.56.120 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.121 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.122 apache_package=httpd php_package=php
```

Make sure to save the `inventory` file and exit.

**Finally**, we still have one more thing to change in our `install_apache.yml` file. In task 2.3, you may notice that the package is assign as `apt`, which will not run in CentOS. Replace the `apt` with `package`. Package is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use `apt`, and for CentOS it will automatically use `dnf`. Make sure to save the file and exit. For more details about the ansible package, you may refer to this documentation: [ansible.builtin.package – Generic OS package manager — Ansible Documentation](https://docs.ansible.com/ansible/latest/builtin/package_module.html)

Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the result.

## Task 2 - Consolidating Playbooks



The screenshot shows a code editor with an Ansible playbook named `install_apache.yml`. The playbook has three tasks: 1. Update repository index for Ubuntu. 2. Install apache2 and php packages for Ubuntu using the `apt` module. 3. Update repository index for CentOS using the `dnf` module. The bottom panel shows the terminal output of the command `ansible-playbook install_apache.yml -K`. The output shows that the playbook was executed on three servers: `server1`, `server2`, and `server3`. The results are summarized in a table below:

| Host    | Task   | OK | Changed | Unreachable | Failed | Skipped | Rescued | Ignored |
|---------|--------|----|---------|-------------|--------|---------|---------|---------|
| server1 | Task 1 | 3  | 1       | 0           | 0      | 3       | 0       | 0       |
| server2 | Task 1 | 3  | 1       | 0           | 0      | 3       | 0       | 0       |
| server3 | Task 1 | 1  | 0       | 0           | 1      | 2       | 0       | 0       |



```
File Edit Selection ... CPE212_jalene
inventory.ini M ! install_apache.yml A, U
inventory.ini
4 server2 ansible_host=192.168.56.103 ansible_user=jalene target_user=jalene
5 server3 ansible_host=192.168.56.108 ansible_user=jalene target_user=jalene
6
7 server1 apache_package=apache2 php_package=libapache2-mod-php_package
8 server2 apache_package=apache2 php_package=libapache2-mod-php_package
9 server3 apache_package=httpd php_package=php
```

```
inventory.ini M ! install_apache.yml A, U
! install_apache.yml
1 ---
2 - hosts: all
3   become: true
4   tasks:
5
6     - name: install apache2 and php packages
7       package:
8         name:
9           - "{{ apache_package }}"
10          - "{{ php_package }}"
11         state: latest
12         update_cache: yes
```

### Supplementary Activity:

1. Create a playbook that could do the previous tasks in Red Hat OS.

```
None
---
- name: Install Apache and PHP on Red Hat systems
  hosts: all
  become: true

  tasks:
    - name: Update repo and install Apache + PHP packages
      ansible.builtin.yum:
        name:
          - httpd
          - php
        state: present
        update_cache: yes
    - name: Enable and start Apache
```

```
ansible.builtin.service:
  name: httpd
  state: started
  enabled: yes
```

This is a sample of a playbook that can do the previous tasks in Red Hat OS. However, I couldn't run or test the playbook because we don't have proper CentOS installed on our PCs.

**Reflections:**

Answer the following:

**1. Why do you think refactoring of playbook codes is important?**

- Refactoring is important because it makes playbooks cleaner, easier to read, and easier to maintain. It helps avoid repeated code and reduces mistakes when updating or reusing playbooks.

**2. When do we use the “when” command in the playbook?**

- We use the when command to run a task only if a certain condition is true. This helps control what tasks run based on things like the system type or installed software.